Evaluation of dried distillers grains as a direct feed and as a protein source in tilapia diets

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Inasmuch as feed accounts for more than 50 percent of production costs for most species, one way to increase production profitability is to lower feed costs. Protein is generally the most expensive dietary component; therefore, determination of less expensive sources of protein that provide good growth would be advantageous for feed producers and farmers alike. Distillers dried grains with solubles (DDGS) are primary fermentation residues from yeast fermentation of cereal grains and are a byproduct of the bourbon and ethanol distilling process. These byproducts are approximately 26-28 percent protein, closely matching the protein requirements of tilapia (25-35 percent according to Lim (1989). Wu *et al.* (1996) evaluated inclusion of DDGS at 30 percent of diet for tilapia and reported good growth. Compared to many species, tilapia require relatively low levels of dietary crude protein and can utilize relatively high levels of plant feedstuffs (Twibell and Brown 1998). This may make much higher inclusion rates possible. Also, Luquet (1991) reported that, in ponds, tilapia utilized meal type feeds as well as pelleted diets, which suggests that DDGS could be fed directly as a loose grain ration, further decreasing feed costs. Two studies were conducted at Kentucky State University to evaluate the potential of DDGS to reduce feed costs in tilapia under practical and controlled conditions.

Table 1. Mean average weight, weight gain (percent increase from initial weight), specific growth rate (SGR), feed conversion ratio (FCR), protein efficiency ratio (PER) and survival of Nile tilapia fed unpelleted distillers grains with solubles (DDGS), steam-pelleted DDGS, or a commercial catfish diet. Values are means ± S.D. of three replicate samples. Means in the same row with different letters are significantly different ($P \leq 0.05$).

Diet	Un-pelleted DDGS	Pelleted DDGS	Commercial Catfish diet
Survival (percent)	97.2±2.8a	96.3±2.3a	95.5±1.3a
Average weight (g)	108.3±6. lb	119.4±3.9b	149.4±6.4a
Average length (cm)	17.8±0.2b	18.4±0. lb	19.7±0.48a
Production (kg/m3)	15.8±0.8b	17.8±0.6b	23.3±1.5a
Weight gain (percent)	416.7±23.9b	459.7±14.7b	574.5±24.7a
Specific growth rate (percent/d)	1.74±0.07c	1,86±0.04b	2.13±0.05a
Feed conversion ratio	2.26±0. 19a	2.07±0.07a	1.66±0.09b
Percent protein deposited	31.70:0.30b	37.34±0.46a	3 5.75±2.85ab
Cost ($)/kg gain'	0.26±0.02c	0.37±0.02b	0.66±0.04a

Cost ($)/kg gain Feed cost ($/kg) X feed conversion ratio

Study 1

In the first study, we evaluated the use of pelleted and unpelleted distillery byproducts as a primary feed for cage reared tilapia. Juvenile tilapia (average individual weight 26.0 g) were obtained from a commercial supplier and stocked into nine 1.0 m^3 cages at 200 fish/cage. Each cage was randomly assigned one of three diets: unpelleted DDGS (25.6 percent protein, 7.3 percent fat); DDGS that had been steam pelleted into 1 mm sinking pellets with 22.5 percent protein, 7.7 percent fat, the lower protein due to dilution by added binder; or a commercial catfish diet[2] (30.7 percent protein, 3.4 percent fat) as a control. There were three replicate cages per treatment. Fish were fed twice daily all they would eat over a 30 min period. All diets were readily consumed.

After 12 weeks, tilapia were harvested and the total number and weight of fish in each cage were determined. Table 1 summarizes the results from the study. Individual weight, individual length, and specific growth rate were significantly higher for fish fed the commercial catfish diet than for fish fed either unpelleted or steam pelleted DDGS. Feed conversion ratio was significantly lower in fish fed the commercial catfish diet (1.7) than for either pelleted (2.1) or unpelleted (2.3) DDGS rations. Specific growth rate and percent protein deposited were significantly higher in fish fed pelleted DDGS than in those fed unpelleted DDGS. However, there was no significant dif-

Table 2. Ingredient composition of practical diets containing 30% distillers grains with solubles (DDGS) in combination with different protein sources and a control. All diets were formulated to contain 32 percent dietary protein.

Ingredients	Control (Lim 1989)	DDGS / Fish	DDGS / M&B	DDGS +Soy
DDGS		30	30	30
Fish Meal	12	8		
Meat and Bone meal			26	
Soybean meal	41	34	16	46
Wheat flour	33.3	16.5	18	11.8
Corn oil	3.3	2,1	1.2	2.5
Fish oil	2.4	1.4	0.8	1.7
Choline chloride	0.5	0.5	0.5	0.5
Mineral mix	0.3	0.3	0.3	0.3
Vitamin mix	0.2	0.2	0.2	0.2
Ascorbic acid	2	2	2	2
Di-calcium phosphate	2	2	2	2
CMC binder	3	3	3	3

Table 3. Average final individual weight, percent survival, feed conversion ratio (FCR), and specific growth rate of tilapia fed experimental diets containing 30 percent distillers dried grains with solubles (DDGS) in combination with different protein sources. Averages are means of 3 replicate aquaria. Numbers within a row followed by different letters were significantly different ($P<0.05$) by ANOVA.

	Control	**DDGS / Fish**	**DDGS / M&B**	**DDGS / Soy**
Final wt(g)	68.la	63.2ab	68.5a	57.8b
Survival (percent)	98.8a	100a	100a	98.8a
FCR	2.lb	2.4b	2.3b	2.7a
SGR(g/day)	0.86a	0.80b	0.87a	0.73c

Table 4. Average relative diet costs, diet costs, and costs of gain including only diet ingredient costs based on Chicago prices[a].

	Control	**DDGS / Fish**	**DDGS / M&B**	**DDGS I Soy**
Relative diet cost	1.0	0.89	0.81	0.83
Diet Cost ($/Kg)	0.37	0.33	0.30	0.31
Cost of gain (s/Kg gain)	0.76b	0.78b	0.69b	0.83a

[a]Reported in Feed stuffs Vol. 73, 53 (Dec. 24, 2001).

ference in average harvest weight, total length, production, weight gain or feed conversion ratio in tilapia when fed DDGS as a loose grain ration or as a steamed pellet. The data agree with those of Luquet (1991), who found that pelleted diets resulted in no conclusive advantage over meal diets in tilapia. Pellet costs increased diet cost 30 percent and only resulted in a 12 percent increase in total production.

Although the commercial catfish diet resulted in the greatest growth, the most economical growth was provided by the DDGS rations. Diets that produce the fastest growth are not always the most profitable. Feed cost per kilogram of gain for tilapia fed the catfish diet ($0.66/kg gain) was significantly higher than for fish fed unpelleted and pelleted DDGS ($0.26/kg gain and $0.37/kg gain, respectively). These figures represent 61 percent and 44 percent decreases in feed cost, respectively, compared to tilapia fed the commercial catfish diet.

Our data suggest that efficient and economical tilapia growth can be obtained using direct feeding of distillery byproducts in situations where optimum growth is not essential. Large reductions (40 to 60 percent) in feed costs should have a positive effect on production economics; however, these savings could be offset by the higher costs of the larger fingerlings that would be required to achieve market sizes within the limited growing season of temperate regions.

Study 2

Due to the high cost, inconsistent supply, and environmental concerns with using fish meal in aquaculture diets, evaluation of alternative protein sources remains a high priority for fish nutritionists. Tilapia diets without fish meal have generally resulted in reduced growth under intensive culture conditions. However, research has shown that combining different animal and plant protein sources can improve utilization. Preliminary studies indicated that DDGS at 30 percent of the diet may be complimentary with other protein sources.

This study was designed to compare diets using different plant and animal protein sources in combination with 30 percent DDGS on growth, feed conversion and production efficiency of tilapia. Twenty-eight juvenile tilapia, with an average individual weight of 2.7g, were stocked into each of 12, 110 l glass aquaria. There were three replicate aquaria per dietary treatment. Experimental diets were based on a model formula by Lim (1989). Each aquarium was randomly assigned one of four diets (Table 2) and fed twice daily to apparent satiation. The system was a shared recycle system to minimize water quality variation among treatments.

After 12 weeks, fish fed the control diet and DDGS + meat and bone meal had higher average weights and specific growth rates than fish fed DDGS + soybean meal (Table 3). Fish fed the DDGS + fish meal diet performed similar to those fed other diets. Feed conversion ratios were similar for fish fed all diets and averaged 2.4. The DDGS + meat and bone meal was the most economical diet, resulting in a cost savings of approximately 20 percent, compared with the control diet which led to no reduction in growth (Table 4).

This study demonstrates that a diet containing 0 percent fish meal, 30 percent DDGS, and 26 percent meat and bone meal can be used for rearing tilapia diets without adverse effects on fish growth. Distillers dried grains with solubles may be complimentary to animal based protein sources, such as meat and bone meal, based on growth similar to diets containing fish meal. The use of DDGS in tilapia diets may allow feed producers more flexibility in formulating nutritious diets at the lowest possible cost by adding another possible ingredient to the least-cost formulation and decreasing the dependence on fish meal and soybean meal.

Notes

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